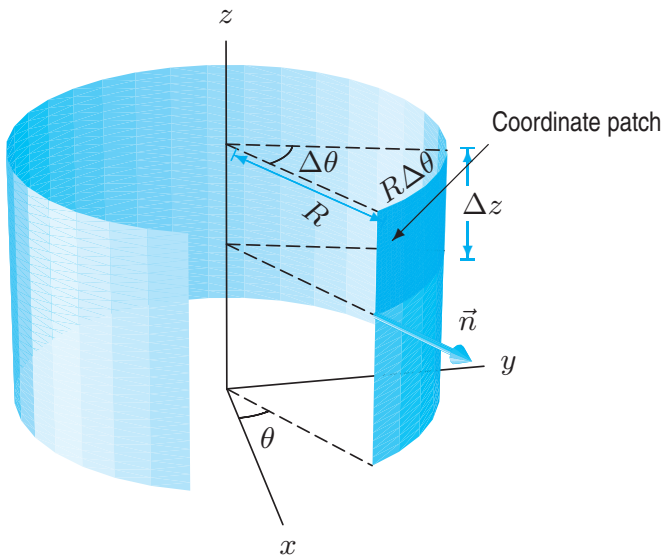
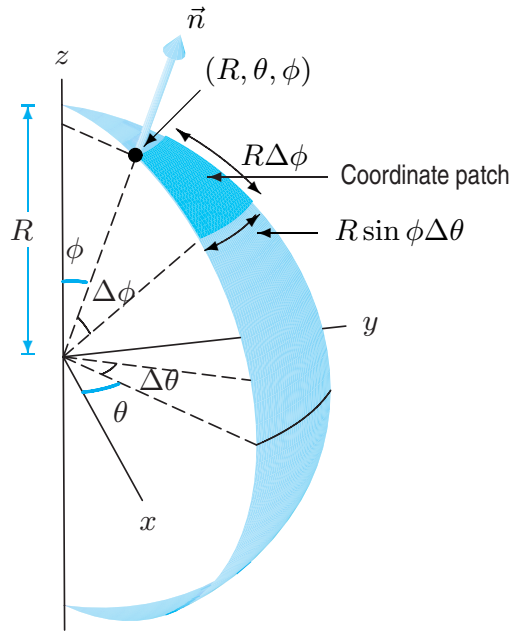
Suggested problems §19.3: 3-9 odd

Suggested problems §19.4: 1-19 odd

Problem 1: Using the diagrams below, compute the vector area element $d\vec{A}$ for cylindrical and spherical surfaces. Assume the cylinder and sphere have radius R.





For the cylinder we have

$$d\vec{A} = \langle \cos(\theta), \sin(\theta), 0 \rangle \; R d\theta dz,$$

while for the sphere we have

$$d\vec{A} = \langle \sin(\phi)\cos(\theta), \sin(\phi)\sin(\theta), \cos(\phi) \rangle \; R^2 \sin(\phi) d\theta d\phi.$$

Problem 2: Compute the flux of

$$\vec{F}(x, y, z) = \langle x, y, 3xz^2 \rangle$$

through the surface S of a cylinder of radius 10, centered on the z-axis, oriented outward, with $0 \leq z \leq 5$.

We have

$$\begin{aligned}
\int_S \langle x, y, 3xz^2 \rangle \cdot d\vec{A} &= \int_0^5 \int_0^{2\pi} \langle 10\cos(\theta), 10\sin(\theta), 30\cos(\theta) \cdot 10^2 \rangle \cdot \langle \cos(\theta), \sin(\theta), 0 \rangle \; 10 d\theta dz \\
&= 100 \int_0^5 \int_0^{2\pi} \; d\theta dz \\
&= 1000\pi.
\end{aligned}$$

Problem 3: Compute the flux of

$$\vec{F}(x, y, z) = \langle 0, 0, z \rangle$$

through the sphere S of radius 5, centered at the origin, and oriented outward.

We have

$$\begin{aligned}\int_S \langle 0, 0, z \rangle \cdot d\vec{A} &= \int_0^{2\pi} \int_0^{\pi} \langle 0, 0, 5\cos(\varphi) \rangle \cdot \langle \sin(\varphi)\cos(\theta), \sin(\varphi)\sin(\theta), \cos(\varphi) \rangle \cdot 25\sin(\varphi) d\varphi d\theta \\ &= 125 \int_0^{2\pi} \int_0^{\pi} \cos^2(\varphi)\sin(\varphi)\ d\varphi d\theta \\ &= \frac{500\pi}{3}.\end{aligned}$$

Problem 4: Compute the divergence $\nabla \cdot \vec{F}$ for the following vector fields.

(a) $\vec{F}(x, y, z) = \langle -x + y, y + z, -z + x \rangle$

$$\nabla \cdot \vec{F} = -1 + 1 - 1 = -1.$$

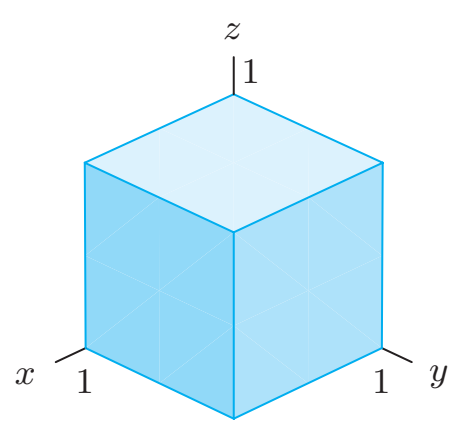
(b) $\vec{F}(x, y, z) = 3x^2\vec{i} - \sin(xz)\vec{j} + 3x^2yz^2\vec{k}$

$$\nabla \cdot \vec{F} = 6x + 6x^2yz.$$

Problem 5: Compute the flux of the vector field

$$\vec{F}(x, y, z) = (x^2 + y^2)\vec{i} + (y^2 + z^2)\vec{j} + (x^2 + z^2)\vec{k}$$

through the surface S of the cube below (oriented outward).



Letting W be the solid cube, by the Divergence Theorem we have

$$\begin{aligned}\int_S \vec{F}\cdot d\vec{A} &= \int_W \nabla\cdot\vec{F}\ dV\\ &= 2\int_0^1\int_0^1\int_0^1 (x+y+z)\ dxdydz\\ &= 3.\end{aligned}$$

Problem 6: Compute the flux of $\vec{F}(x,y,z) = \langle xy, yz, xz\rangle$ through the sphere S of radius 1 centered at the origin (oriented outward).

We let W be the solid sphere whose boundary is S. Then the Divergence Theorem gives us

$$\begin{aligned}\int_S \vec{F}\cdot d\vec{A} &= \int_W \nabla\cdot\vec{F}\ dV\\ &= \int_W (x+y+z)\ dV\\ &= \int_0^\pi\int_0^{2\pi}\int_0^1 (\rho\sin(\varphi)\sin(\theta)+\rho\cos(\varphi)+\rho\sin(\varphi)\cos(\theta))\rho^2\sin(\varphi)\ d\rho d\theta d\varphi\\ &= \frac{1}{4}\int_0^\pi\int_0^{2\pi}(\sin^2(\varphi)\sin(\theta)+\sin(\varphi)\cos(\varphi)+\sin^2(\varphi)\cos(\theta))\ d\theta d\varphi\\ &= \frac{\pi}{2}\int_0^\pi \sin(\varphi)\cos(\varphi)\ d\varphi\\ &= 0.\end{aligned}$$

Problem 7: Let S be an outward-oriented closed cylinder of height 2 with its base a disk of radius 2 in the xy-plane centered at the origin. Compute the flux of $\vec{F}(x,y,z) = \langle 0, y, 0\rangle$ through S in two different ways.

We first compute the flux directly as $\int_S \vec{F}\cdot d\vec{A}$. Since the cylinder has a top and bottom, we have to break the flux integral into three integrals:

$$\begin{aligned}\int_S \vec{F}\cdot d\vec{A} &= \int_{\text{cylinder wall}} \vec{F}\cdot d\vec{A} + \int_{\text{top}} \vec{F}\cdot d\vec{A} + \int_{\text{bottom}} \vec{F}\cdot d\vec{A}\\ &= 4\int_0^2\int_0^{2\pi}\langle 0,\sin(\theta),0\rangle\cdot\langle\cos(\theta),\sin(\theta),0\rangle\ d\theta dz + \int_{\text{top}} \vec{F}\cdot d\vec{A} + \int_{\text{bottom}} \vec{F}\cdot d\vec{A}\\ &= 4\int_0^2\int_0^{2\pi}\sin^2(\theta)\ d\theta dz + \int_{\text{top}} \vec{F}\cdot d\vec{A} + \int_{\text{bottom}} \vec{F}\cdot d\vec{A}\\ &= 8\pi + \int_{\text{top}} \vec{F}\cdot d\vec{A} + \int_{\text{bottom}} \vec{F}\cdot d\vec{A}\\ &= 8\pi + \iint_{\text{disk}} \langle 0,y,0\rangle\cdot\langle 0,0,1\rangle\ dxdy + \iint_{\text{disk}} \langle 0,y,0\rangle\cdot\langle 0,0,-1\rangle\ dxdy\\ &= 8\pi.\end{aligned}$$

We now compute the flux by using the Divergence Theorem. Let W be the solid cylinder of which S is the boundary. Then

$$\int_W \nabla\cdot\vec{F}\ dV = \int_W\ dV = \pi\cdot 2^2\cdot 2 = 8\pi.$$